Group Assignment 2 Instructions

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1 The story

Your team works for a short-term-rental analytics start-up. Your first two clients are in **city A** and **city B**. For them you build a lakehouse over Airbnb data, an exploratory dashboard, a Genie Agent, and a model that suggests a price for a listing. The model goes to production.

One week later, sales signs a third client in **city C**, in the same region. "Your pipeline takes the city as a parameter, right? Can we have prices there by Friday?"

You run what you have on city C — unchanged — and see what happens. Then you fix it, and report honestly: what broke, what you changed in the pipeline, what you changed in the model, and how much worse the predictions are than in the cities you trained on.

The point of the assignment is that last report. A pipeline that only works on the data it was built on is a prototype; the third city is where you find out which one you have.

2 Data and regions

The data is from Inside Airbnb (CC BY 4.0). For each city use the **latest snapshot** and the detailed `listings.csv.gz`; state the snapshot date of every city in your README. Other files (`calendar.csv.gz`, `reviews.csv.gz`) are optional.

Each team works in one **region**. All cities in a region share a currency, so prices can be compared without conversion.

1. Claim a region in the Teams channel: `<team name>: <region>`. Each region can be claimed by one team only.
2. From your region, choose **two training cities (A and B)** and **one new-client city (C)**, and post them in the same thread. Say in one sentence why you chose C.
3. Do not explore city C until your pipeline and model for A and B are finished. City C is your unseen data; the assignment only works if it stays unseen.

#	Region	Currency	Cities
1	California	USD	Los Angeles, San Diego, San Francisco, Santa Clara County
2	Texas	USD	Austin, Dallas, Fort Worth
3	US Northeast	USD	New York City, Boston, Jersey City, Rhode Island
4	US South	USD	Broward County, Nashville, New Orleans, Asheville
5	Canada	CAD	Toronto, Montreal, Vancouver, Victoria
6	Australia	AUD	Sydney, Melbourne, Brisbane
7	United Kingdom	GBP	London, Greater Manchester, Edinburgh, Bristol
8	Northern Italy	EUR	Milan, Florence, Venice, Bologna
9	Southern Italy	EUR	Rome, Naples, Sicily, Puglia
10	Spain	EUR	Madrid, Barcelona, Valencia, Sevilla, Málaga
11	France	EUR	Paris, Bordeaux, Lyon, Pays Basque
12	Benelux	EUR	Amsterdam, Brussels, Antwerp
13	Germany & Austria	EUR	Berlin, Vienna, Munich
14	Switzerland	CHF	Vaud, Zurich, Geneva

Some entries are whole regions or islands rather than cities (Sicily, Puglia, Pays Basque). Choosing one of them as city C makes the transfer harder, and that is a legitimate choice — say so in your report.

All work runs on Databricks Free Edition and must cost nothing.

3 Deliverables

3.1 Exploratory analysis

A separate, reproducible notebook on cities A and B (Lecture 6), with:

- a summary section at the top: what you now know about the data, and what you still do not;
- a description of the dataset: snapshot dates, grain (what one row is), record counts per city;
- a profile of the columns you use: types, missing-value rates and their *pattern*, distributions of price and availability, outliers and how you classify them;
- the cleaning and transformations the pipeline must apply, each with its reason;
- a **leakage review**: which columns must not be used to predict price, and why. Look carefully — some columns are computed from the price.

3.2 Solution architecture

A diagram and a short description of your lakehouse. Use the medallion architecture (Lecture 4): raw files in bronze, cleaned and typed listings in silver, metrics, model features and predictions in gold.

3.3 ETL pipeline

- A Databricks Job or a declarative pipeline (SQL or Python) that loads, cleans and transforms the data.
- **One pipeline with the city as a parameter**, run for each city in sequence or in parallel. Adding a city must not mean copying code.

3.4 Price model

A model that predicts the `price` of a listing in cities A and B. Machine learning on Spark is covered in Lecture 9; use what you already know, with scikit-learn or Spark ML.

- **Target:** the `price` column — the price per night the listing advertised on the snapshot date, in the region's currency. Predicting log price and converting back is allowed.
- **Split:** hold out a test set (for example 20% of listings, per city) before any model choice. Every number you report comes from that test set.
- **Baseline:** the median price for the same room type and number of guests (`accommodates`), computed on the training data. Your model must be compared against it.
- **Metric:** mean absolute error (MAE) in the region's currency, reported for city A, city B and both together. You may add other metrics.
- **Output:** predictions for every listing written to a gold table, so the dashboard and Genie can use them.

A simple model that clearly beats the baseline, with features you can explain, is better than a complex one you cannot.

3.5 Dashboard

An AI/BI dashboard (Lecture 8) with at least three pages:

- **City overview** — per city: listings, median price, availability, review scores, hosts, room-type mix, neighbourhoods by price.
- **Cross-city comparison** — A against B: side-by-side metrics and the differences that matter.
- **Model** — predicted against actual price, error by room type or neighbourhood, and the listings the model under- or over-prices most. After the city C run, add city C here.

The examples are a starting point, not a checklist. Every page should have a title that states a finding (Lecture 7).

3.6 Genie Agent

A Genie Agent (formerly called a Genie space) over your gold tables, configured with instructions and example questions, for example:

- "What is the median price of entire homes in City A?"
- "Which neighbourhoods in City B have the most listings?"
- "Where does the model under-price listings the most?"

3.7 The new client: city C

When the pipeline and model for A and B are finished:

1. Run the **unchanged** pipeline and model on city C, and write down what happened: did it run, what failed, what came out, and the MAE on city C.
2. Fix what needs fixing. Keep a list of every change and its reason, and sort each one into one of two kinds:
    - **pipeline fixes** — the pipeline should have handled this from the start (an unexpected value, a schema difference, a failed download);
    - **model changes** — the model did not transfer (a feature that means something different in city C, a price level the model never saw).
3. Report three numbers side by side: the baseline on city C, the model's MAE on A and B, and the model's MAE on city C (first run, and after your changes, if they changed the model). Explain the gap.

A large first-run error that you can explain is worth more than a small one you cannot.

3.8 Presentation

- the task and your solution at a high level;
- analysis summary, architecture and pipeline;
- the model: features, baseline against model, test-set error;
- **the city C story**: what happened on the first run, what you changed and why, and the error on city C;
- a demo of the dashboard and Genie, and the tricks you used to configure Genie;
- challenges, and each team member's contribution.

3.9 Repository

The repository holds the notebooks, pipeline definition, a README with the region, the three cities and their snapshot dates, and the presentation as a PDF or a link to it.

4 Timeline

The assignment takes two weeks.

Week	Milestone
1	Region and cities claimed; data loaded; exploratory analysis and leakage review Pipeline and model for cities A and B finished;
2	Dashboard and Genie; City C: first run, fixes, report; presentation

Do not start week 2 early. The first run on city C must be with the pipeline and model as they were at the end of week 1.

5 Evaluation

Deliverable	What we look for
Exploratory analysis	Reproducible; missingness and outliers explained, not just counted; a leakage review that finds the leaks
Architecture and pipeline	Medallion layers with a clear purpose; one parameterised pipeline that really runs per city
Price model	Honest test-set evaluation against the baseline; features you can justify; no leakage
Dashboard and Genie	Pages answer real questions, with message titles; Genie answers the example questions correctly
City C	An honest first-run result; changes listed and classified; the gap between A/B and C explained
Presentation	One story, told clearly; the demo works

6 Submission

Submit the link to the repository with the deliverables. Only the team admin submits.

Each team member can earn up to **10 points**.